THE ONE EXTRA CHECK

Logarithmic inequalities are solved with the same algebraic steps as equations, with one critical addition: the base decides whether the inequality direction is kept or reversed.

LOGARITHMIC INEQUALITY

$$\log_a(x) > c$$

split by **the base** a

↑

CASE 1

$a > 1$

Direction preserved

the logarithm is increasing: larger arguments produce larger outputs

E.G. $\log_2(x) > 3 \Rightarrow x > 2^3 = 8$

↓

CASE 2

$0 < a < 1$

Direction reversed

the logarithm is decreasing: larger arguments produce smaller outputs

E.G. $\log_{1/2}(x) > 3 \Rightarrow x < (1/2)^3 = 1/8$

TAKEAWAY

Domain restrictions apply in both cases: every argument must stay positive, and that constraint intersects with the algebraic solution.